

DD-Robocon 2026 – Stage I Submission

- Robot 1 (R1) is a manual robot.
- Robot 2 (R2) is a fully autonomous robot.

Section 1: Solution Ideas

Robot 1 (R1)

1. Overall Dimensions and Estimated Weight

The overall height, length and breath are **1000mm, 1000mm** and **800mm** respectively . The weight of R1 is **51.69kg** after including all the components , material used, motors etc .

2. Drive Mechanism

We have used a four-wheel drive mechanism, in which wheels are **mecanum wheels** and we have used **planetary gear box brushless motor** with this a **belt mechanism**. We are using mecanum wheels because Move **forward / backward**, Move **sideways (strafing)**, Move **diagonally** **Rotate in place**, Do all of these simultaneously .

3. Type of Actuators

We are using brushless motors with planetary gear box with 600rpm and 72 kg-cm torque , 12V DC 0.6A 7.5W Solenoid Lock , Linear Actuator Stroke Length 300MM, 7mm/S, 1500N, 12V, Linear Actuator Stroke Length 500MM 7mm/S 1000N 12V , Waveshare 30KG Serial Bus Servo , TowerPro MG996R Digital High Torque Servo Motor (360° Rotation)

4. Type of sensors integrated

MT6835 magnetic encoder module PWM/SPI brushless motor encoder with magnet , ultrasonic sensors

Robot 2 (R2)

1. Overall Dimensions and Estimated Weight

The overall height, length and breath are **1000mm, 1000mm** and **800mm** respectively . The weight of R2 is **65 kg** after including all the components , material used, motors etc .

2. Drive Mechanism

We have used a four-wheel drive mechanism, in which wheels reference is taken by the **Synchro-Linkage** mechanism, developed by **Professor Eric Laithwaite** .

2. Type of Actuators Used

We have used **4 brushless motors** **with planetary gear box** . the reason behind using brushless motors is it give **more torque** than brushed one's , 12V 150MM Stroke Length Linear Actuator 6000N 5mm/s .

3. Type of Sensors Integrated

MT6835 magnetic encoder module PWM/SPI brushless motor 21BIT encoder with magnet, Ultrasonic Distance sensor, Sensor, Jetson Orin, 9-dof IMU 1Mono, USB Camera D435, Depth camera , RP Lidar SE2 .

Section 2: Design Document, CAD Model and Mechanism Explanation

1. Description of Robot 1 (R1) with CAD view

1.1 Functional overview of R1

R1 is capable of : 1. Navigating the arena and R1 pathway with high stability and precision.

2. its gripper can hold staff with ease. 3. mecanum wheels helps **R1** to operate in all direction easily .

4. **R1** can pick up 1 kung-fu scroll at a time and it can store 2 scrolls inside its body .

5. **R1** can carry assembled weapons inside its body and also take it out easily.

6. **R1** is capable of lifting R2 to top most row of tic tac toe to help it place kung-fu scrolls or to push opponent teams scrolls .

7. **R1** is capable of operating on a ground with inclination of 15 degrees.

1.2 Design philosophy

1.2.1 Material used: we have used Aluminium Extrusion because of following reasons Aluminium extrusions provide

an excellent strength-to-weight ratio, reducing the overall mass of the robot while maintaining sufficient structural rigidity. This improves acceleration, manoeuvrability, and energy efficiency. The T-slot profile allows easy assembly, disassembly, and reconfiguration without welding , Dimensional Accuracy and Alignment , Corrosion Resistance, Flexibility for Mounting Components, Cost-Effective.

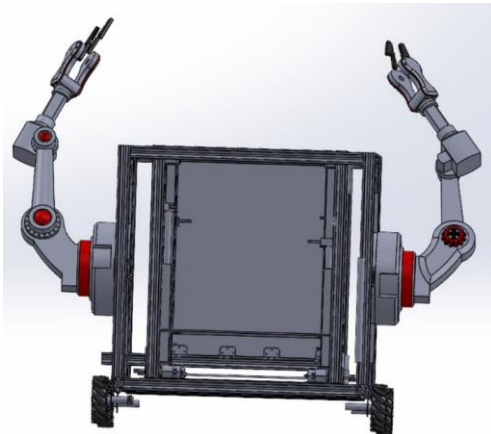
1.2.2 Robotic arm : The robotic arm used in R1 and R2 are same . both of them have 5 joints including 1 shoulder and a circular gripper is used .

1.2.3 Wheels : we have used mecanum wheels because of this R1 can Move **forward / backward**, Move **sideways (strafing)**, Move **diagonally Rotate in place**, Do all of these **simultaneously** .

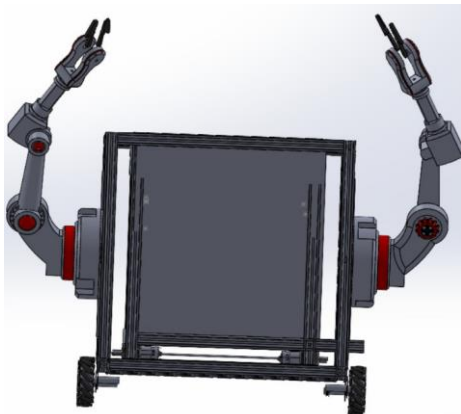
1.2.4 Belt mechanism : we have used belt mechanism because Belts naturally absorb vibrations and sudden load variations, protecting motors and gearboxes from impact loads during rapid acceleration, deceleration or collisions. belt mechanisms operate with minimal frictional losses , cost efficient and can work in confine spaces .

1.3 CAD views

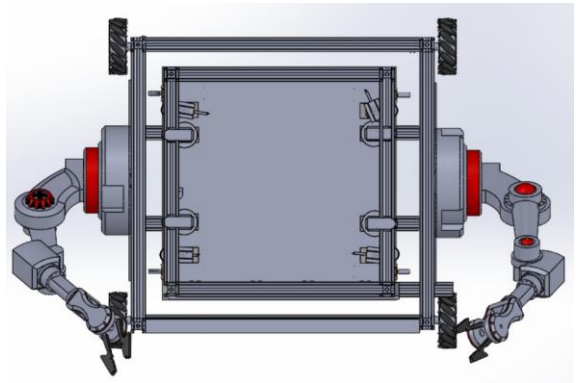
1.3.1 Front view



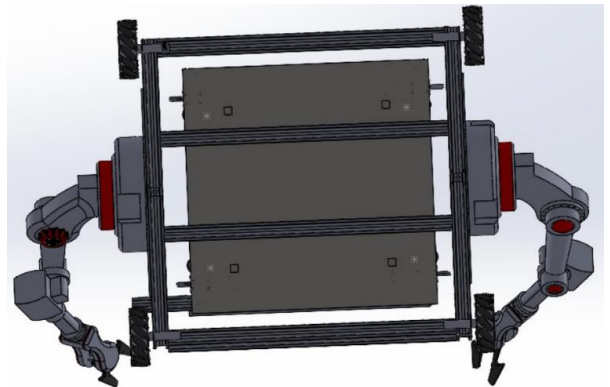
1.3.2 Back



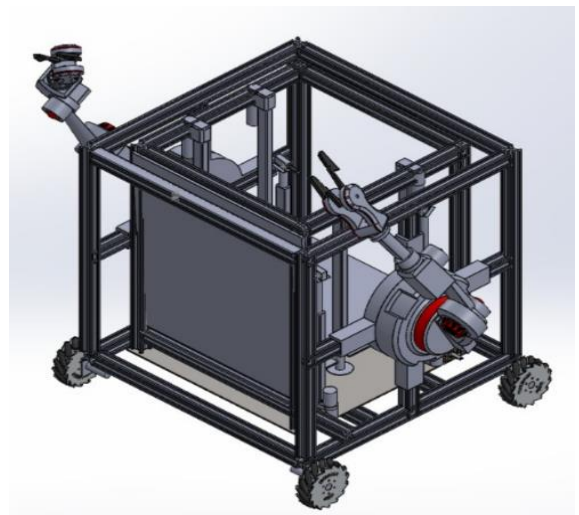
1.3.3 Top



1.3.4 Bottom



1.3.5 Isometric



1.3.6 Complete assembly

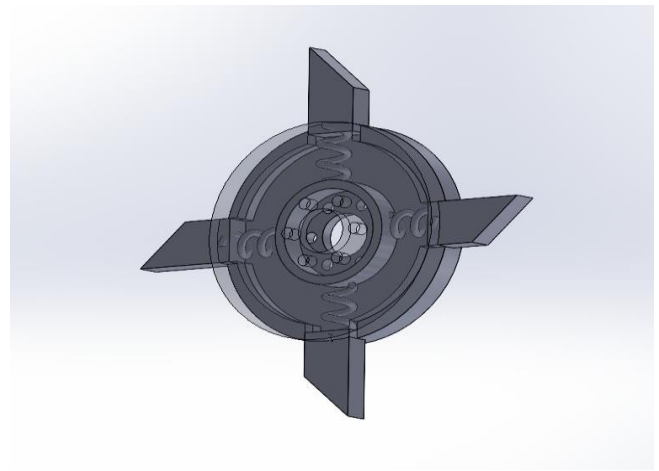
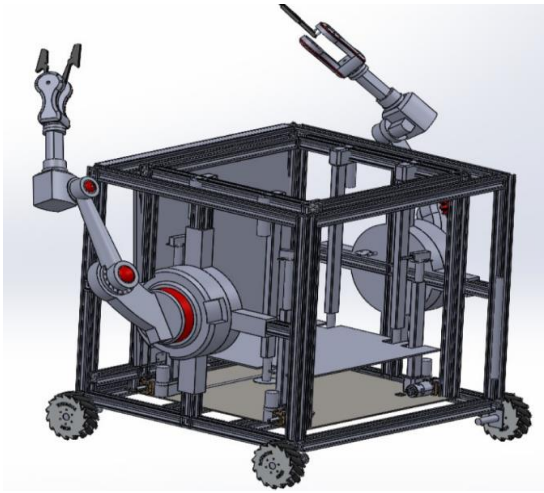


Figure1:synchro-linkage wheels

2. Description of Robot 2 (R2) with CAD Views

1.4 Functional overview of R2

1. R2 is capable of operating on flat ground as well as on forest with ease

2. It can pick spearhead and hold them till assembly.

3. R2 can store 2 assembled weapons inside its body and also two kung-fu scrolls and take them out whenever required.

4. Its arm are capable of placing objects and pushing opponent teams scrolls from tic tac toe

5. R2 is capable of operating on a ground with inclination of 15 degrees.

6. R2 is capable of identification/differentiating between fake, R1 and R2 scrolls.

2.1 Design philosophy

2.2.1 Material used: 1 material used: we have used Aluminium Extrusion because of following reasons Aluminium extrusions provide an excellent strength-to-weight ratio, reducing the overall mass of the robot while maintaining sufficient structural rigidity. This improves acceleration, manoeuvrability, and energy efficiency. The T-slot profile allows easy assembly, disassembly, and reconfiguration without welding, Dimensional Accuracy and Alignment, Corrosion Resistance, Flexibility for Mounting Components, Cost-Effective.

2.2.2 Robotic arm : The robotic arm used in R1 and R2 are same. both of them have 5 joints including 1 shoulder and a circular gripper is used.

2.2.3 Wheels : we have used Synchro-Linkage wheels. How does wheels work : 1. When the machine is on flat ground, the whole triangle doesn't spin. Instead, the two bottom wheels just roll normally, like a regular cart. The third wheel sits on top, basically "resting."

2. when wheels are about to touch stairs then front wheel hits the vertical face of a step, it can't roll forward anymore. The motor keeps pushing, and since the bottom wheel is stuck, the entire triangle assembly is forced to rotate.

3. On the flat surface, the outer wheel touches the ground first and the inner wheel which has triangle assembly is floating in air but the triangles coming out wheel hits the ground at certain angle and it goes in, after this these wheels work smooth on flat ground.

2.2.4 Belt mechanism: belt mechanism is same R1 2.2.5 Lifting Kungfu scrolls inside R2 :We are using 4 linear actuators and above a aluminium plate is kept and all this is kept inside a cuboidal box of metal sheets and Al extrusions and when needed the linear actuators lift the platform upwards and with that scrolls will come out in arena.

2.2 CAD views

3. Game Object Pick-and-Place Mechanism (R1/R2)

3.1 Mechanism Description and Animation Views

We are using 2 robotic arms each bot. arm consist of 4 links, 1 Shoulder, 1 gripper. we are using circular gripper which are capable of picking up spearheads in a single arm. when R2 has picked 1 spear head then it holds the spearheads top towards its body, holding it in such a way that head touches body of R2 perpendicularly by this there is less chance of spear head slipping from gripper. meanwhile R1 pick up staff from staff rack and then a weapon is assembled and can be stored in R1/R2. when R1/R2 are about to pick up kung-fu scrolls both arms rotate 90 degrees and the area of arm facing towards scroll has rubber grips on the surface. then both arms close down towards scroll just like a human is picking a big box with both hands, when scroll is picked by R1/R2 they can store scrolls inside their body. R1 can store 2 scrolls and R2 Can store 1 scroll. when R1/R2 are in arena, we have used a mechanism of linear actuators that can lift scrolls from their body.

3.2 Calculations / Justifications

1.Height of spear to be picked up = 50cm + 5.25cm = 55.25cm, Total = 60cm arm length , Lowest height of reach = **55.25 cm**,Highest height of reach = **100 cm**,

J₂: 45.81 × 2 = 91 kg-cm (joint 2)

J₃: 21.06 × 2 = 42 kg-cm(joint 3)

J₄ - J₆: 15 kg-cm (Wrist, joint 4)

τ = 24.57 kg-cm (torque required for base of robotic arm)

these are the calculated results and the complete calculation's word file has been attached to drive .

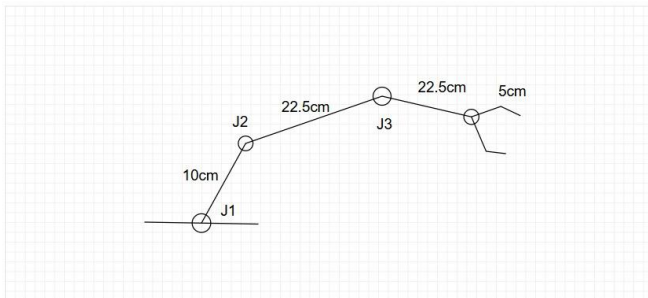


figure 3 : rough sketch of robotic arm used

3.3 CAD Model Views

4. Object Detection and Autonomous Navigation (R2)

4.1 System Description and Animation Views

4.1.1 Object Detection System: Multi-Camera Hardware Array: The vision system utilizes three standard RGB cameras mounted on the sides for a wide, peripheral field of view, alongside one front-facing depth camera. This configuration ensures comprehensive visual coverage of the arena with zero blind spots.

Edge Processing: The video streams from all four cameras are synchronized and fed into the central edge computing unit (NVIDIA Jetson Orin Nano). This setup handles the high bandwidth of multiple simultaneous video feeds while maintaining the real-time processing speeds necessary for competition.

4.1.2 Algorithm: Deep Learning Object Detection: A custom-trained deep learning model is deployed to identify and classify the target boxes. The model processes the 2D RGB streams from the side cameras for rapid bounding box generation and object classification.

Sensor Fusion for 3D Localization: When a target box enters the front depth camera's field of view, the algorithm fuses the 2D bounding box detections with the spatial depth point cloud. This sensor fusion extracts the exact x, y, z coordinates of the box relative to the robot, enabling precise mechanical alignment and interaction.

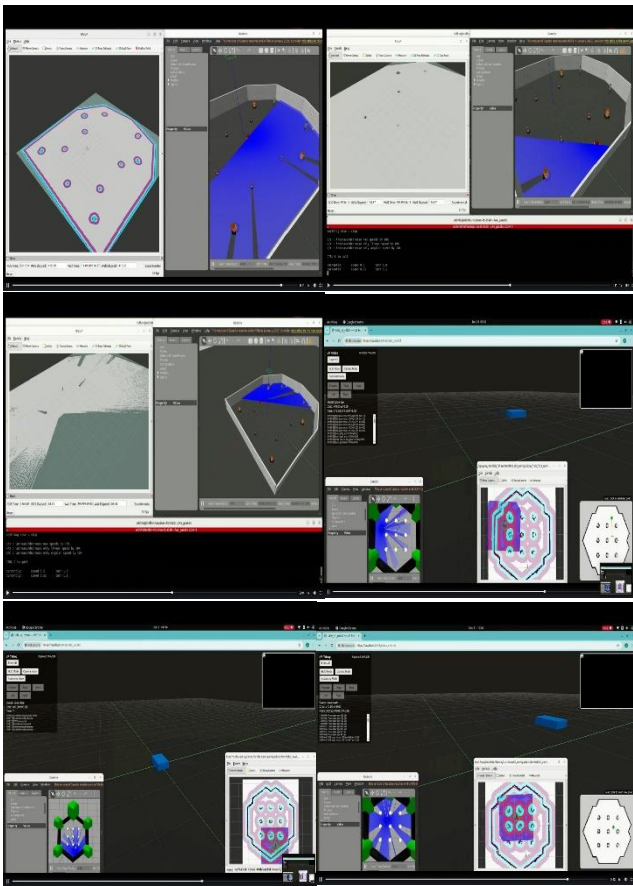
4.2 Autonomous Navigation

4.2.1 Navigation Algorithm and Software Environment: Odometry & Sensor Fusion (EKF): The robot calculates its

continuous local position using a combination of Wheel Encoders and an IMU (Inertial Measurement Unit). The wheel encoders provide raw distance measurements, while the IMU provides high-frequency data on orientation (yaw) and acceleration. These inputs are mathematically fused using an Extended Kalman Filter (EKF) via the ROS 2 Jazzy robot_localization package to actively cancel out physical wheel slip and sensor drift. High-Fidelity Global Mapping: While the EKF handles local movement, the environment is perceived globally using the RPLIDAR S2E. Its Direct Time-of-Flight (DToF) technology and 32kHz sampling rate match real-time Lidar scans against the static arena map, correcting any accumulated odometry errors and maintaining exact global positioning.

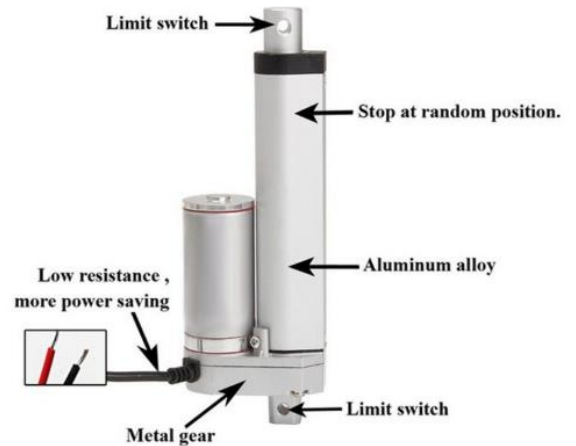
Software Stack & Short-Goal Routing: Autonomous navigation is driven by the Nav2 stack, which processes the fused EKF odometry and Lidar data to update global and local costmaps in real-time. The algorithm is optimized using a sequential waypoint strategy where the global planner sends a series of "short goals" straight through the middle of the arena. This approach keeps the robot centralized, reduces computational overhead, and allows the local trajectory planner to react instantly to dynamic obstacles.

4.2.2 Simulation Animation Views: Vision System Validation: An animation view (simulated in Gazebo/RViz) illustrating the overlapping field-of-view cones of the four cameras, demonstrating the deep learning model successfully recognizing target boxes from multiple angles simultaneously. Dynamic Navigation & Sensor Fusion Simulation: A top-down RViz animation displaying the local costmap generated by the RPLIDAR S2E. The view highlights the Nav2 behaviour tree in action, visualizing the robot smoothly executing its path from one short goal to the next. Crucially, the animation shows the raw encoder path diverging slightly, while the fused EKF odometry path remains perfectly aligned with the Lidar map, proving the stability of the navigation system.



5.3 CAD Model Views During Lifting

6. Weapon Assembly Mechanism



6.1 Mechanism Description

We have used circular grippers with a width wide enough to perfectly hold the Quick couplers of the spearhead and the staff as well.

The R2 bot will follow its designated path towards the spearhead rack and will pick up the desired spearhead with its gripper mechanism. The bot is programmed to itself rotate the spearhead and make it parallel towards its body, holding it firmly against its body so that it does not slips. Now the R1 bot will be moved towards the R2 bot manually to attach the staff to the spearhead to make a complete weapon. There is an LED constructed on the R1 manual bot which gets turned on as soon as the weapon is completed. This ensures that the R2's gripper leaves the assembled weapon for its further usage.

6.2 Github Links:-

Contents:-

1. Views of R1
2. Views of R2
3. Lifting mechanism video
4. Calculation of linear actuator
5. Calculation of robotic arm
6. Object detection and navigation

<https://github.com/ShivamMaurya14/Robocon-Team-CEAR>

5. Lifting Mechanism

5.1 Mechanism Description and Animation Views:

We have used a total of 8 linear actuators in which 4 gives 350N , have 300mm stroke length and 4 give 1000N , have 500mm stroke length . 4 actuators [lets call it as A2(1000N)]are attached upside down and 4 actuators [lets call them A1(350N)] are attached normally (at 90 degrees)to the base plate . We have used another steel plate (P1) of 5mm thickness. We have attached four solenoid locks to each A1's limit switches (refer to figure 12). A plate has been attached to A2's limit switch of radius = 25mm. We have used four circular steel plates that are attached to the limit switch of A2, which are holding plate P1 above them.

Originally, the P1 rests on the four circular plates. The A2 actuator starts pulling plate P1 upwards, when it reaches to the solenoid lock present on A1's limit switch, then the solenoid lock will get activated and the rod inside comes out from lock, meanwhile the A2's limit switch hits it max height that it can pull P1 plate. Now, the P1 plate gets support from the rod which has came out of the A1's solenoid lock. Now, the A1 actuators get activated and they will take the P1 plate to the maximum height (750mm from base plate).

5.2 Calculations / Justifications

No of actuators used : 8

350N , 300mm stroke length : 4

1000N , 500mm stroke length : 4

Total mass of R2 = 47 kg